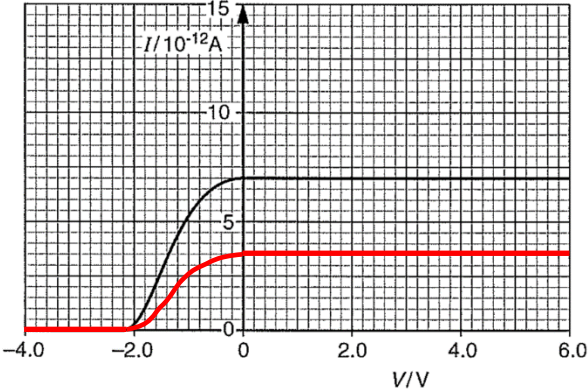


7ai	As p.d. becomes negative, there are photoelectrons that reach collector C, contributing to the photocurrent, as they have <u>kinetic energy greater than the work done against electric field</u> between the metal plates.(As the p.d. becomes more negative, only the more energetic photoelectrons reach the collector C) Hence, the sloping section of the graph shows that photoelectrons are emitted from metal plate E with <u>a range of kinetic energies</u>.	B1 B1
7aii	From Fig. 7.2, the stopping potential, $V = -2.2 \text{ V}$ (Since stopping potential stops the most energetic photoelectrons emitted from metal plate E, by conservation of energy) loss in kinetic energy of most energetic photoelectrons = work done against e-field $\frac{1}{2} m v_{\max}^2 - 0 = qV$ $v_{\max} = \sqrt{\frac{qV}{\frac{1}{2} m}} = \sqrt{\frac{(-1.60 \times 10^{-19})(-2.2)}{\frac{1}{2}(9.11 \times 10^{-31})}}$ $= 8.79 \times 10^5 \text{ m s}^{-1}$	B1 M1 A1

7aiii	 frequency is constant → <u>stopping potential is the same</u> intensity is halved → rate of photons incident on metal plate E is halved → rate of photoelectrons emitted is halved → <u>photocurrent is halved</u> at every respective value of p.d.	B1 B1
7b	Since $E = hf$ or hc/λ, from work function energy, $\text{threshold wavelength} = \frac{hc}{\text{work function energy}} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{5.8 \times 10^{-19}} = 343 \text{ nm}$ (For photoelectrons to be emitted, the wavelength of the radiation should be less than the threshold wavelength.) Since the wavelength of the radiation from the second lamp is longer than the threshold wavelength [appropriate comment <u>comparing</u> wavelengths/ energies/ frequencies], there will be <u>no effect</u> on photoelectric current (as this wavelength does not cause photoelectric effect of zinc). Note: Also accept analysis using threshold frequency or minimum energy for photoelectron emission.	C1 A1 A1

8a	Fast moving/ high speed/ high (kinetic) energy electrons moving towards the target	B1
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